

Cryptography and Network Security

Quadratic Congruence

- In cryptography, we often encounter equations where variables are raised to powers, like quadratic equations.
- A general quadratic congruence looks like:

$$a_2x^2 + a_1x + a_0 \equiv 0 \pmod{n}$$

- But for simplicity and cryptographic relevance, we focus on:

$$\boxed{x^2 \equiv a \pmod{n}.}$$
 where $a_2 = 1$ and $a_1 = 0$.

Quadratic Congruence Modulo a Prime

- We want to find the solutions for an equation of the form as above, in which 'n' is prime 'p', a is an integer such that p does not divide a.

- **Equation:**

$$x^2 \equiv a \pmod{p}$$

- **Theorem:** This equation has either:
 - No solutions, or
 - Exactly two incongruent solutions modulo p .
- If x is a solution, then $-x \pmod{p}$ is also a solution, and these are the only two if they exist.

Example The equation $x^2 \equiv 3 \pmod{11}$

We check squares modulo 11 systematically:

Step 1 – Try all residues from 0 to 10:

$0^2 = 0 \rightarrow \text{remainder } 0$	$4^2 = 16 \rightarrow \text{remainder } 5$	
$1^2 = 1 \rightarrow \text{remainder } 1$	$5^2 = 25 \rightarrow \text{remainder } 3$	✓ FOUND! $x \equiv 5$
$2^2 = 4 \rightarrow \text{remainder } 4$	$6^2 = 36 \rightarrow \text{remainder } 3$	✓ FOUND! $x \equiv 6$
$3^2 = 9 \rightarrow \text{remainder } 9$		

$7^2 = 49 \rightarrow \text{remainder } 5$
$8^2 = 64 \rightarrow \text{remainder } 9$
$9^2 = 81 \rightarrow \text{remainder } 4$
$10^2 = 100 \rightarrow \text{remainder } 1$

Step 2 – Solutions found:

$$x \equiv 5 \pmod{11} \text{ or } x \equiv 6 \pmod{11}$$

Step 3 – Why two solutions?

If r is a solution, then $p - r$ is also a solution because $(p - r)^2 \equiv r^2 \pmod{p}$.

Here: $11 - 5 = 6 \rightarrow$ so 6 is automatically a solution if 5 works.

Example The equation $x^2 \equiv 2 \pmod{11}$

Step 1 – Check all squares modulo 11:

From previous calculations, squares mod 11 are:

$$0, 1, 4, 9, 5, 3, 3, 5, 9, 4, 1$$

(corresponding to $x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$).

Step 2 – Observation:

The number **2** never appears in the list of squares modulo 11.

Step 3 – Conclusion:

No integer x satisfies

$$x^2 \equiv 2 \pmod{11}$$

Practice Exercise

Exercise 1: Solve the congruence (or determine if no solution exists):

$$x^2 \equiv 4 \pmod{13}$$

Exercise 2

Solve the congruence (or determine if no solution exists):

$$x^2 \equiv 7 \pmod{13}$$

Exercise 3

Solve the congruence (or determine if no solution exists):

$$x^2 \equiv 3 \pmod{7}$$

Quadratic Residues and Nonresidues

- If $x^2 \equiv a \pmod{p}$ has **two solutions**, then a is called a **Quadratic Residue (QR)** of p .
- If $x^2 \equiv a \pmod{p}$ has **no solutions**, then a is called a **Quadratic Nonresidue (QNR)** of p .

As discussed earlier, for $p = 11$:

3 is a QR (we found solutions: 5 and 6)

2 is a QNR (no solutions exist)

Important Theorem:

In Z_p^* the set $\{1, 2, \dots, p-1\}$, exactly **half** are QRs and **half** are QNRs.

For prime p :

$$\text{Number of QR} = \frac{p-1}{2}, \text{Number of QNR} = \frac{p-1}{2}$$

Example for $p = 11$:

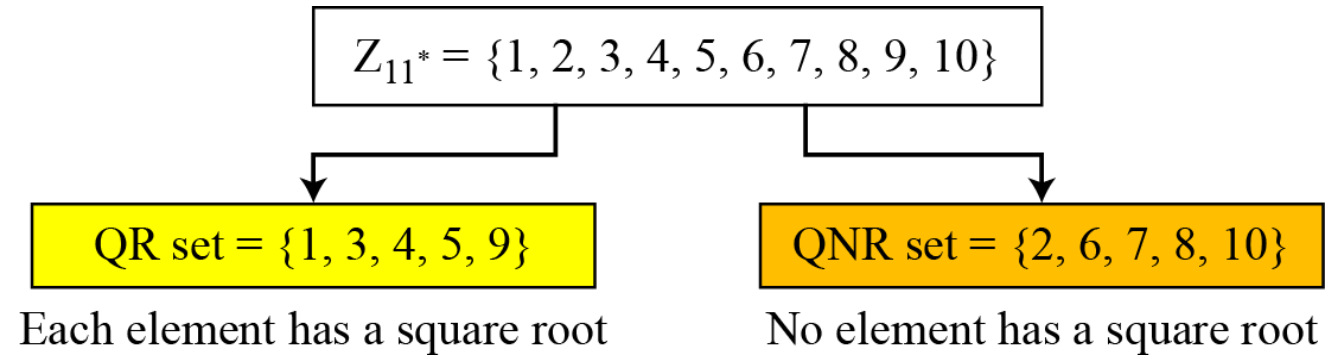
$p-1 = 10$ elements in Z_{11}^*

$\frac{10}{2} = 5$ QRs and 5 QNRs

Example

There are 10 elements in $\mathbb{Z}_{11}^*$. Exactly five of them are quadratic residues and five of them are nonresidues. In other words, $\mathbb{Z}_{11}^*$ is divided into two separate sets, QR and QNR, as shown in Figure.

Figure *Division of $\mathbb{Z}_{11}^*$ elements into QRs and QNRs*



Steps:

1. What is $\mathbb{Z}_{11}^*$?

- Numbers modulo 11 without 0:

$$\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

- All have multiplicative inverses mod 11.

2. Quadratic Residue (QR) Definition

- A number a is a **QR** mod 11 if there exists x such that:

$$x^2 \equiv a \pmod{11}$$

- Example:** 3 is QR because $5^2 \equiv 3 \pmod{11}$.

3. Find All QRs mod 11

Squaring each element:

x	$x^2 \bmod 11$
1	1
2	4
3	9
4	5
5	3
6	3
7	5
8	9
9	4
10	1

Distinct results $\rightarrow$ QR set:

$$\boxed{\{1, 3, 4, 5, 9\}}$$

4. Quadratic Non-Residues (QNRs)

What's left in $\mathbb{Z}_{11}^*$:

$$\boxed{\{2, 6, 7, 8, 10\}}$$

No square root mod 11.

5. Why Half-and-Half?

For any odd prime p :

- $\mathbb{Z}_p^*$ has $p - 1$ elements.
- Each QR comes from **two** different numbers' squares.
- Number of QRs = $\frac{p-1}{2}$.

For $p = 11$:

$$\frac{11 - 1}{2} = 5 \text{ QRs, } 5 \text{ QNRs.}$$

6. Visual Division

$$\begin{aligned}\mathbb{Z}_{11}^* &= \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \\ QR &= \{1, 3, 4, 5, 9\} \text{ (has square root)} \\ QNR &= \{2, 6, 7, 8, 10\} \text{ (no square root)}\end{aligned}$$

Exercise

Exercise 1

For the group $\mathbb{Z}_{13}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$:

Compute the square of each element modulo 13.

(That is, find $x^2 \bmod 13$ for $x = 1, 2, \dots, 12$).

List all distinct quadratic residues (QR) in $\mathbb{Z}_{13}^*$.

List all quadratic non-residues (QNR) in $\mathbb{Z}_{13}^*$.

Verify that exactly half of the elements are QR and half are QNR.

Exercise 2

For the group $\mathbb{Z}_{17}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$:

Compute the square of each element modulo 17 (only check $x = 1, 2, \dots, 8$ if you use the symmetry).

List all distinct quadratic residues (QR) in $\mathbb{Z}_{17}^*$.

List all quadratic non-residues (QNR) in $\mathbb{Z}_{17}^*$.

Verify that exactly half of the elements are QR and half are QNR.

Euler's Criterion

How can we check to see if an integer is QR modulo P ?

- For large primes p , checking all squares is impractical. **Euler's Criterion** gives us a direct test.

Euler's Criterion Theorem

For an odd prime p and a not divisible by p ($a \in \mathbb{Z}_p^*$):

$$a^{(p-1)/2} \bmod p = \begin{cases} 1 & \Rightarrow a \text{ is a QR modulo } p \\ -1 & \Rightarrow a \text{ is a QNR modulo } p \end{cases}$$

Key Insight: The exponent $\frac{p-1}{2}$ acts as a "QR detector."

Why This Works?

From **Fermat's Little Theorem**:

$$a^{p-1} \equiv 1 \pmod{p} \text{ for any } a \not\equiv 0 \pmod{p}$$

Taking square root of both sides:

$$\left(a^{(p-1)/2}\right)^2 \equiv 1 \pmod{p}$$

$$\text{So } a^{(p-1)/2} \equiv \pm 1 \pmod{p}.$$

• If a is a QR ($a \equiv x^2$), then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$

• If a is a QNR, it must be -1

Example

Find out if 14 is a QR in $\mathbb{Z}_{23}^*$:

Test: Is 14 a QR modulo 23?

Step 1: Apply Euler's Formula

$$14^{(23-1)/2} = 14^{11} \pmod{23}$$

Step 2: Compute Step-by-Step

We break the exponent 11 into binary: $11 = 8 + 2 + 1$

We compute powers of 14 modulo 23:

$$1. 14^1 = 14$$

$$2. 14^2 = 196 \equiv 12 \pmod{23} \\ (196 \div 23 = 8 \text{ remainder } 12)$$

$$3. 14^4 = (14^2)^2 \equiv 12^2 = 144 \equiv 6 \pmod{23} \\ (144 \div 23 = 6 \text{ remainder } 6)$$

$$4. 14^8 = (14^4)^2 \equiv 6^2 = 36 \equiv 13 \pmod{23}$$

Step 3: Combine for 14^{11}

$$14^{11} = 14^8 \times 14^2 \times 14^1 \\ = 13 \times 12 \times 14 \pmod{23}$$

First multiplication:

$$13 \times 12 = 156 \\ 156 \div 23 = 6 \text{ remainder } 18 \Rightarrow 156 \\ \equiv 18 \pmod{23}$$

Second multiplication:

$$18 \times 14 = 252 \\ 252 \div 23 = 10 \text{ remainder } 22 \Rightarrow 252 \\ \equiv 22 \pmod{23}$$

So:

$$14^{11} \equiv 22 \pmod{23}$$

Step 4: Interpret Result

$$22 \pmod{23} \equiv -1 \pmod{23}$$

Since the result is -1 modulo 23, by Euler's Criterion:
14 is a Quadratic Non-Residue (QNR) modulo 23.

Example

Test: Is 16 a QR modulo 23?

Step 1: Apply Formula

$$16^{(23-1)/2} = 16^{11} \pmod{23}$$

Step 2: Compute

We break the exponent 11 into powers of 2: $11 = 8 + 2 + 1$

1. $16^1 = 16$
2. $16^2 = 256 \equiv 3 \pmod{23}$
3. $16^4 = 3^2 = 9 \pmod{23}$
4. $16^8 = 9^2 = 81 \equiv 12 \pmod{23}$

Step 3: Combine for 16^{11}

$$\begin{aligned} 16^{11} &= 16^8 \times 16^2 \times 16^1 \\ &= 12 \times 3 \times 16 \pmod{23} \end{aligned}$$

$$12 \times 3 = 36 \equiv 13 \pmod{23}$$

$$13 \times 16 = 208 \equiv 1 \pmod{23}$$

Step 4: Interpret Result

$$16^{11} \equiv 1 \pmod{23}$$

Since result = 1 $\rightarrow$ **16 is a QR modulo 23**

(Verification: $4^2 = 16 \pmod{23}$)

Exercise

Exercise 1

Determine whether **8** is a quadratic residue modulo **17** using Euler's Criterion.

Exercise 2

Determine whether **10** is a quadratic residue modulo **19** using Euler's Criterion.

Quadratic Equation Modulo a Prime

What Euler's Criterion Can't Do

Euler's Criterion only tells us **whether** $x^2 \equiv a \pmod{p}$ has a solution, **not what the solution is**.

Example:

For $a = 16$, $p = 23$, as shown earlier, 16 is a QR mod 23, but Euler's Criterion doesn't give $x = 4$ or $x = 19$ as solutions.

Prime Types: $p = 4k + 1$ vs $p = 4k + 3$

All odd primes fall into one of two families:

$$p = 4k + 1$$

Example: $13 = 4 \cdot 3 + 1$, $17 = 4 \cdot 4 + 1$

Solving $x^2 \equiv a$ can be done using **Tonelli-Shanks algorithm**

More computationally complex

$$p = 4k + 3$$

Example: $7 = 4 \cdot 1 + 3$, $23 = 4 \cdot 5 + 3$

Also called **Blum primes** (when q is also $4k + 3$, $p = 2q + 1$ etc.)

Quadratic equations have **direct formula solution**

Why Focus on $p = 4k + 3$?

In the **Rabin Cryptosystem** (a public-key cryptosystem):

- We choose p, q both of form $4k + 3$
- This ensures **efficient decryption** using the direct formula
- Security is based on hardness of factoring $n = p \cdot q$

Solving Quadratic Equation Modulo a Prime

Special Case: $p = 4k + 3$. If p is in the form $4k + 3$, $p \equiv 3 \pmod{4}$ and a is a QR in $\mathbb{Z}$, then

$$x \equiv a^{(p+1)/4} \pmod{p} \quad \text{and} \quad x \equiv -a^{(p+1)/4} \pmod{p}$$

Example

Solve the following quadratic equations: $x^2 \equiv 3 \pmod{23}$

Solutions:

Step 1: Check if p is of the form $4k + 3$

$$23 = 4 \times 5 + 3 \text{ so } k = 5, p \equiv 3 \pmod{4}$$

The special formula applies:

$$x \equiv \pm a^{(p+1)/4} \pmod{p}$$

where $a = 3$, $p = 23$, so $\frac{p+1}{4} = \frac{24}{4} = 6$.

Step 2: Check if 3 is a QR modulo 23 using Euler's Criterion

Euler's Criterion: $3^{(23-1)/2} \equiv 3^{11} \pmod{23}$.

We compute $3^{11} \pmod{23}$ via binary exponentiation:

1. $3^1 \equiv 3$
2. $3^2 = 9$
3. $3^4 \equiv 9^2 = 81 \equiv 81 - 3 \times 23 = 81 - 69 = 12$
4. $3^8 \equiv 12^2 = 144 \equiv 144 - 6 \times 23 = 144 - 138 = 6$
5. $3^{11} = 3^8 \times 3^2 \times 3^1 \equiv 6 \times 9 \times 3$

First: $6 \times 9 = 54 \equiv 54 - 2 \times 23 = 54 - 46 = 8$

Then: $8 \times 3 = 24 \equiv 24 - 23 = 1$

So: $3^{11} \equiv 1 \pmod{23}$

✓ By Euler's Criterion: 1 means **3 is a QR modulo 23**.

Step 3: Apply the formula $x \equiv \pm a^{(p+1)/4} \pmod{p}$

We need $3^{(23+1)/4} = 3^6 \pmod{23}$.

Compute 3^6 :

1. $3^2 = 9$
2. $3^4 \equiv 9^2 = 81 \equiv 12 \pmod{23}$ (as before)
3. $3^6 = 3^4 \times 3^2 \equiv 12 \times 9 = 108$

$$108 \div 23 = 4 \text{ remainder } 108 - 92 = 16$$

So:

$$3^6 \equiv 16 \pmod{23}$$

Step 4: Write solutions

$$x \equiv 16 \pmod{23} \text{ or } x \equiv -16 \equiv 7 \pmod{23}$$

✓ Check:

$$16^2 = 256, 256 \div 23 = 11 \text{ remainder } 256 - 253 = 3 \checkmark$$

$$7^2 = 49, 49 \div 23 = 2 \text{ remainder } 49 - 46 = 3 \checkmark$$

Final Answer:

$x \equiv 7 \text{ or } 16 \pmod{23}$

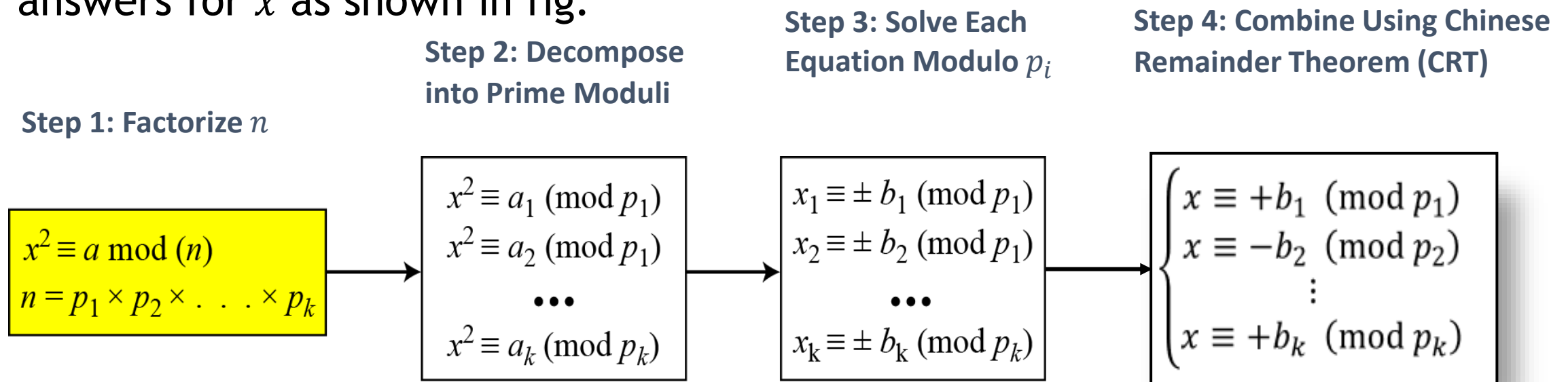
Exercise

Exercise 1: Solve the following quadratic equations: $x^2 \equiv 2 \pmod{11}$

Exercise 2: Solve the following quadratic equations: $x^2 \equiv 7 \pmod{19}$

Solving Quadratic Equation Modulo a Composite

- It can be done by solving a set of congruence modulo a prime.
- In other words, we can decompose $x^2 \equiv a \pmod{n}$ if we have the factorization of n .
- Now we can solve each decomposed equation (if solvable) and find k pairs of answers for x as shown in fig.



Step 5: Total Solutions

We get 2^k distinct solutions for x modulo n .

Example Solve the equation $x^2 \equiv 36 \pmod{77}$.

Step 1: Factorize n

$$n = 77 = 7 \times 11$$

So $p_1 = 7$, $p_2 = 11$, and $k = 2$.

Step 2: Decompose into prime moduli

We have:

$$x^2 \equiv 36 \pmod{7}$$

$$x^2 \equiv 36 \pmod{11}$$

Simplify:

$$36 \bmod 7 = 1$$

$$36 \bmod 11 = 3$$

So:

$$x^2 \equiv 1 \pmod{7}$$

$$x^2 \equiv 3 \pmod{11}$$

Check if 1 is QR mod 7 and 3 is QR mod 11

Step 3: Solve each equation modulo p_i

For $p_1 = 7$: Apply Euler's Criterion

$$x^2 \equiv 1 \pmod{7}$$

$$x \equiv \pm a^{(p+1)/4} \pmod{p}$$

Here $a = 1$, $p = 7$: $x \equiv \pm 1^{(7+1)/4} \pmod{7}$

$$x \equiv 1 \pmod{7} \text{ or } x \equiv -1 \equiv 6 \pmod{7}$$

So $b_1 = 1$ and $-1 \equiv 6 \pmod{7}$.

For $p_2 = 11$: Apply Eulers Criterion

$$x^2 \equiv 3 \pmod{11}$$

$$x \equiv \pm a^{(p+1)/4} \pmod{p}$$

Here $a = 3$, $p = 11$

$$x \equiv \pm 3^{(11+1)/4} \pmod{11}$$

$$x \equiv \pm 3^3 \pmod{11}$$

$$x \equiv \pm 5$$

So:

$$x \equiv 5 \pmod{11} \text{ or } x \equiv 6 \pmod{11}$$

So : $b_2 = 5, 6 \equiv -5 \pmod{11}$.

Step 4: Combine using CRT

We have 4 combinations:

$$x \equiv 1 \pmod{7} \text{ and } x \equiv 5 \pmod{11}$$

$$x \equiv 1 \pmod{7} \text{ and } x \equiv 6 \pmod{11}$$

$$x \equiv 6 \pmod{7} \text{ and } x \equiv 5 \pmod{11}$$

$$x \equiv 6 \pmod{7} \text{ and } x \equiv 6 \pmod{11}$$

Step 5: Form four systems of congruences

System 1:

$$\begin{cases} x \equiv 1 \pmod{7} \\ x \equiv 5 \pmod{11} \end{cases}$$

System 2:

$$\begin{cases} x \equiv 1 \pmod{7} \\ x \equiv 6 \pmod{11} \end{cases}$$

System 3:

$$\begin{cases} x \equiv 6 \pmod{7} \\ x \equiv 5 \pmod{11} \end{cases}$$

System 4:

$$\begin{cases} x \equiv 6 \pmod{7} \\ x \equiv 6 \pmod{11} \end{cases}$$

Step 6: Solve using Chinese Remainder Theorem (CRT)

System 1: $x \equiv 1 \pmod{7}, x \equiv 5 \pmod{11}$

$$m_1=7, a_1=1; m_2=11, a_2=5$$

They are coprime: $\gcd(7,11) = 1$

Step 2: Compute M

$$M = m_1 \cdot m_2 = 7 \times 11 = 77$$

Step 3: Compute M_i

$$M_1 = \frac{M}{m_1} = \frac{77}{7} = 11$$

$$M_2 = \frac{M}{m_2} = \frac{77}{11} = 7$$

Step 4: Find modular inverses y_i

We need y_1 such that $M_1 y_1 \equiv 1 \pmod{m_1}$:

$$11y_1 \equiv 1 \pmod{7}$$

$11 \equiv 4 \pmod{7}$, so equation becomes:

$$4y_1 \equiv 1 \pmod{7}$$

Inverse of 4 modulo 7: $4 \times 2 = 8 \equiv 1 \pmod{7} \rightarrow y_1 = 2$.

Similarly, y_2 such that $M_2 y_2 \equiv 1 \pmod{m_2}$:

$$7y_2 \equiv 1 \pmod{11}$$

Inverse of 7 modulo 11: $7 \times 8 = 56 \equiv 1 \pmod{11} \rightarrow y_2 = 8$.

Step 5: Apply CRT formula

The solution is:

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{M}$$

Substitute:

$$x \equiv (1)(11)(2) + (5)(7)(8) \pmod{77}$$

$$x \equiv 22 + 280 \pmod{77}$$

$$x \equiv 302 \pmod{77}$$

Step 6: Reduce modulo 77

$$77 \times 3 = 231, 302 - 231 = 71.$$

So:

$$x \equiv \mathbf{71 \pmod{77}}$$

System 2: $x \equiv 1 \pmod{7}, x \equiv 6 \pmod{11}$

Step 1: Define moduli and residues

$$m_1 = 7, a_1 = 1$$

$$m_2 = 11, a_2 = 6$$

$$\gcd(7, 11) = 1 \text{ (coprime)}$$

Step 2: Compute M

$$M = m_1 \cdot m_2 = 7 \times 11 = 77$$

Step 3: Compute M_i

$$M_1 = \frac{M}{m_1} = \frac{77}{7} = 11$$

$$M_2 = \frac{M}{m_2} = \frac{77}{11} = 7$$

Step 4: Find modular inverses y_i

We already found from the previous system:

For $M_1 = 11$ modulo $m_1 = 7$:

$$11 \equiv 4 \pmod{7}$$

$$4y_1 \equiv 1 \pmod{7} \rightarrow y_1 = 2 \text{ since } 4 \times 2 = 8 \equiv 1.$$

For $M_2 = 7$ modulo $m_2 = 11$:

$$7y_2 \equiv 1 \pmod{11} \rightarrow y_2 = 8 \text{ since } 7 \times 8 = 56 \equiv 1$$

Step 5: Apply CRT formula

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{77}$$

$$x \equiv (1)(11)(2) + (6)(7)(8) \pmod{77}$$

$$x \equiv 22 + 336 \pmod{77}$$

$$x \equiv 358 \pmod{77}$$

Step 6: Reduce modulo 77

$$77 \times 4 = 308$$

$$358 - 308 = 50$$

So:

$$x \equiv \mathbf{50 \pmod{77}}$$

System 3: $x \equiv 6 \pmod{7}, x \equiv 5 \pmod{11}$

Step 1: Define moduli and residues

$$m_1 = 7, a_1 = 6$$

$$m_2 = 11, a_2 = 5$$

$$\gcd(7, 11) = 1$$

Step 2: Compute M

$$M = m_1 \cdot m_2 = 7 \times 11 = 77$$

Step 3: Compute M_i

$$M_1 = \frac{M}{m_1} = \frac{77}{7} = 11$$

$$M_2 = \frac{M}{m_2} = \frac{77}{11} = 7$$

Step 4: Modular inverses y_i

From before:

$$M_1 = 11, m_1 = 7 : 11 \equiv 4 \pmod{7}, 4y_1 \equiv 1 \pmod{7} \rightarrow y_1 = 2.$$

$$M_2 = 7, m_2 = 11 : 7y_2 \equiv 1 \pmod{11} \rightarrow y_2 = 8.$$

Step 5: Apply CRT formula

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{77}$$

$$x \equiv (6)(11)(2) + (5)(7)(8) \pmod{77}$$

$$x \equiv 132 + 280 \pmod{77}$$

$$x \equiv 412 \pmod{77}$$

Step 6: Reduce modulo 77

$$77 \times 5 = 385$$

$$412 - 385 = 27$$

So:

$$x \equiv 27 \pmod{77}$$

System 4: $x \equiv 6 \pmod{7}, x \equiv 6 \pmod{11}$

Step 1: Define moduli and residues

$$m_1 = 7, a_1 = 6$$

$$m_2 = 11, a_2 = 6$$

.

Step 2: Compute M

$$M = m_1 \cdot m_2 = 7 \times 11 = 77$$

Step 3: Check for a simpler observation

If $x \equiv 6 \pmod{7}$ and $x \equiv 6 \pmod{11}$,
then $x - 6$ is divisible by both 7 and 11.

Since 7 and 11 are coprime, $x - 6$ is divisible by $7 \times 11 = 77$.

Thus:

$$x \equiv 6 \pmod{77}$$

No need for full CRT formula here — the solution is direct.

Step 4: Verify with CRT formula anyway for consistency

From before:

$$M_1 = 11, M_2 = 7$$

$$y_1 = 2, y_2 = 8$$

CRT formula:

$$x \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod{77}$$

$$x \equiv (6)(11)(2) + (6)(7)(8) \pmod{77}$$

$$x \equiv 132 + 336 \pmod{77}$$

$$x \equiv 468 \pmod{77}$$

$$77 \times 6 = 462$$

$$468 - 462 = 6$$

So:

$$x \equiv 6 \pmod{77}$$

Total solutions

We have $k = 2$ primes, so $2^k = 4$ distinct solutions modulo 77:

$$x \equiv 6, 27, 50, 71 \pmod{77}$$

Check:

$$6^2 = 36 \checkmark$$

$$27^2 = 729 \equiv 729 - 9 \times 77 = 729 - 693 = 36 \checkmark$$

$$50^2 = 2500 \equiv 2500 - 32 \times 77 = 2500 - 2464 = 36 \checkmark$$

$$71^2 = 5041 \equiv 5041 - 65 \times 77 = 5041 - 5005 = 36 \checkmark$$

Final Answer:

6,27,50,71

All mod 77.

Complexity: How hard is it to solve a quadratic congruence modulo a composite?

The main task is the factorization of the modulus. In other words, the **complexity of solving a quadratic congruence modulo a composite is the same as factorizing a composite integer**. As we have seen, if n is very large, factorization is infeasible.

Solving a quadratic congruence modulo a composite is as hard as factorization of the modulus.

Exercise

Problem 1

$$x^2 \equiv 64 \pmod{91}$$

Hint: $91 = 7 \times 13$

Problem 2

$$x^2 \equiv 49 \pmod{143}$$

Hint: $143 = 11 \times 13$